



90-Day Java + Backend SQL Bootcamp

From Zero to Interview-Ready Java Backend Developer
2 hrs/day · 90 Days · Beginner → Job-Ready

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1. Reusable Templates

Daily Homework Standard

- ☐ Solve assigned Java problems (see Practice Standard)
- ☐ Solve assigned SQL queries
- ☐ Review today's notes
- ☐ Update GitHub if working on a project

Daily Revision Routine (final 10 min of session)

Review today's concepts · Update flashcards · Revisit yesterday's mistakes

Practice Standard

Weekday	Difficulty	Focus
Mon	Easy	New concept intro
Tue	Easy–Medium	Reinforcement
Wed	Medium	Applied problems
Thu	Medium	Applied problems
Fri	Hard	Challenge set
Sat	Project	Build/ship
Sun	Revision + Quiz	Weekly Checkpoint

Weekly Checkpoint

(every Revision + Quiz day)

Revision Complete · Java Practice Complete · SQL Practice Complete · Quiz Finished · Project Updated · GitHub Updated

Project Completion Checklist

(every project's final day)

Code Completed · README Updated · SQL Script Saved · GitHub Updated · Tested · Documentation Complete

Monthly Assessment Template

- ☐ Java Coding Test (that month's topics)
- ☐ SQL Test (that month's topics)
- ☐ Project Review
- ☐ Mock Interview (Month 2+)
- ☐ Weak Topic Review

2. Master Roadmap

Wk	Days	Java Focus	SQL Focus	Project (Stage Days)
1	1–7	Variables, Operators, Conditionals, Switch	SELECT, WHERE, ORDER BY, LIKE/IN/BETWEEN	—
2	8–14	Loops, Nested Loops, Break/Continue, Loop Algorithms	GROUP BY, HAVING, Aggregates	Calculator (13 build, 14 refactor)
3	15–21	Arrays (1D/2D), Array Algorithms, Strings, StringBuilder	INNER/LEFT/RIGHT/FULL/Self JOIN, Multi-table Joins	—
4	22–28	Methods, Overloading, Recursion	Subqueries, Correlated Subqueries	Student Mgmt (25–27) · Month 1 Assessment (28)
5	29–35	Classes, Constructors, Encapsulation, static, Packages	CTEs, Recursive CTEs, Views, Indexes	—
6	36–42	Inheritance, super, Overriding, Polymorphism, Abstraction	Stored Procedures, Window Fns, Transactions/ACID	Bank Mgmt (41–42)
7	43–49	Interfaces, Collections, Comparable/Comparator	Normalization (1NF–BCNF), Denormalization	Bank finish (44) · Hospital DB (47–48)
8	50–56	Exceptions, File Handling, Multithreading	Query Optimization	Library Mgmt (54) · E-commerce DB (53–55) · Month 2 Assessment (56)
9	57–63	DSA: Searching, Sorting, 2D Arrays, Strings, Stack	Execution Plans, DB Design, ER Diagrams	E-commerce finish (61) · Bank DB (62–63)
10	64–70	DSA: Linked Lists, HashMap, BST, Traversals	Performance Tuning	Bank DB finish (66) · Employee Mgmt (69)
11	71–77	DSA: AVL, Graphs, BFS/DFS, Recursion, Backtracking	Capstone DB build	Mock Coding + SQL (77)
12	78–84	Inventory System, Interview Qs	SQL Interview Qs, Resume Prep	Inventory (78–79) · Mock Interview 1 (84)
13	85–90	Capstone Backend, Full Mock Interview	Capstone integration, SQL Mock	Capstone (85–88) · Graduation (90)

Unless marked, every day = Java Session → SQL Session → Homework Standard → Revision Routine.

3. Weekly Session Tables

Week 1 (Days 1–7) — Fundamentals

Day	Java Session	SQL Session
1	Variables & Data Types	RDBMS Basics, SELECT
2	Operators — Calculator	WHERE Basics
3	Logical/Bitwise Operators	WHERE + AND/OR/NOT
4	Scanner Input	LIKE / IN / BETWEEN
5	if-else — Grade Calculator	ORDER BY
6	switch-case — Menu Program	ORDER BY + LIMIT/Alias
7	Revision + Quiz	Revision + Quiz

Week 2 (Days 8–14) — Loops & Aggregation

Day	Java Session	SQL Session
8	for Loop — Pattern Printing	GROUP BY
9	while/do-while — Sum & Average	HAVING
10	Nested Loops — Multiplication Table	COUNT, SUM
11	break & continue	AVG, MIN, MAX
12	Loop Algorithms — Prime/Factorial/Fibonacci	GROUP BY + HAVING Combined
13	Calculator — Build	calc_history Table
14	Revision + Quiz — Refactor Calculator	Revision + Quiz — Timed Aggregate Quiz

Week 3 (Days 15–21) — Arrays, Strings & Joins

Day	Java Session	SQL Session
15	1D Arrays — Sum/Max/Min	INNER JOIN
16	2D Arrays — Matrix	LEFT JOIN
17	Array Algorithms — Reverse/Search/Sort	RIGHT JOIN
18	String Basics & Immutability	FULL JOIN
19	String Methods — Palindrome/Anagram	Self Join

Day	Java Session	SQL Session
20	StringBuilder	Multi-table Joins (3+)
21	Revision + Quiz — Array+String Mix	Revision + Quiz — Timed Join Quiz

Week 4 (Days 22–28) — Methods, Recursion, Subqueries + Project

Day	Java Session	SQL Session
22	Methods & Parameters — Utility Methods	Mixed Join Practice
23	Method Overloading	Subqueries — Intro
24	Recursion Basics — Factorial/Fibonacci	Correlated Subqueries
25	Student Mgmt — Design Classes	Design students/marks Tables
26	Student Mgmt — CRUD Logic	CRUD SQL
27	Revision — Polish Project	Revision — Optimize Queries
28	Month 1 Assessment — Coding Test	Month 1 Assessment — SQL Test

Week 5 (Days 29–35) — OOP Foundations + CTE/Views/Indexes

Day	Java Session	SQL Session
29	Classes & Objects	CTE Basics
30	Constructors & this	Recursive CTEs
31	Encapsulation & Access Modifiers	Views
32	Immutable Class Design	Views — Update Limitations
33	static Keyword	Indexes — Intro
34	Packages & import	Index Practice — Composite
35	Revision + Quiz — Mixed OOP	Revision + Quiz — CTE/View/Index Quiz

Week 6 (Days 36–42) — Inheritance, Polymorphism + Stored Proc/Window Fn + Project

Day	Java Session	SQL Session
36	Inheritance & extends	Stored Procedures — Intro
37	super Keyword	Stored Procedures — Parameters
38	Method Overriding	Window Fns — ROW_NUMBER/RANK
39	Polymorphism (Runtime)	Window Fns — LEAD/LAG

Day	Java Session	SQL Session
40	Abstraction & Abstract Classes	Transactions & ACID
41	Bank Mgmt — Account Classes	Design accounts/transactions
42	Bank Mgmt — Deposit/Withdraw Logic + Revision	Transaction-safe Transfer Query + Revision

Week 7 (Days 43–49) — Interfaces, Collections + Normalization + Project

Day	Java Session	SQL Session
43	Interfaces	Normalization — 1NF/2NF
44	Bank Mgmt — Finish & Polish	Normalization — 3NF/BCNF
45	Collections — List	Denormalization Trade-offs
46	Collections — Set	Mixed Normalization Practice
47	Collections — Map	Hospital DB — Design Schema
48	Comparable & Comparator	Hospital DB — Finish + Queries
49	Revision + Quiz — Mixed Collections	Revision + Quiz — Normalization/Design Quiz

Week 8 (Days 50–56) — Exceptions, Files, Threads + Optimization + Project + Assessment

Day	Java Session	SQL Session
50	try-catch-finally	Query Optimization — Intro
51	Custom Exceptions — throw/throws	Optimization — Avoiding SELECT *
52	File Handling — Read/Write	Mixed Advanced Practice
53	Multithreading Basics	E-commerce DB — Design Schema
54	Library Mgmt — Build	E-commerce DB — Continue
55	Revision — Polish Month 2 Projects	Revision — Optimize E-commerce Queries
56	Month 2 Assessment — OOP+Collections Test	Month 2 Assessment — Advanced SQL Test

Week 9 (Days 57–63) — DSA: Arrays/Strings/Stack/Queue + DB Design/ER

Day	Java Session	SQL Session
57	DSA — Searching (Linear/Binary)	Execution Plans (EXPLAIN)

Day	Java Session	SQL Session
58	DSA — Sorting (Bubble/Selection/Insertion)	Execution Plan Practice
59	DSA — 2D Array Problems	Database Design Principles
60	DSA — String Pattern Problems	ER Diagram — Basics
61	DSA — Advanced String Problems	ER Diagram Practice + Finish E-commerce DB
62	DSA — Stack	Bank DB Project — Design
63	Revision + Quiz — DSA Queue	Revision + Quiz — Bank DB Continue

Week 10 (Days 64–70) — DSA: Linked List/HashMap/Trees + Performance Tuning + Project

Day	Java Session	SQL Session
64	DSA — Singly Linked List	Performance Tuning — Basics
65	DSA — Doubly/Circular Linked List	Performance Tuning Practice
66	DSA — HashMap Internals	Bank DB Project — Finish
67	DSA — Trees: BST Basics	Capstone DB — Requirements
68	DSA — Tree Traversals	Capstone DB — Schema Draft v1
69	Employee Mgmt — Build	Capstone DB — Refine Schema
70	Revision + Quiz — Linked List/Tree Mix	Revision + Quiz — Performance-Tuning Quiz

Week 11 (Days 71–77) — DSA: Advanced Trees/Graphs/Backtracking + Capstone DB

Day	Java Session	SQL Session
71	DSA — AVL Intuition	Capstone DB — Seed Data
72	DSA — Graph Representation	Capstone DB — Analytical Queries
73	DSA — BFS	Capstone DB — Analytical Queries 2
74	DSA — DFS	Capstone DB — Indexing Pass
75	DSA — Recursion Advanced	Capstone DB — Finalize & Document
76	DSA — Backtracking Basics (N-Queens)	SQL Interview Qs — Joins/Subqueries
77	Mock — Sudoku Solver + Coding Test	Mock — SQL Interview

Week 12 (Days 78–84) — Java Project + Interview Prep

Day	Java Session	SQL Session
78	Inventory System — Design Classes	SQL Interview Qs — Window Functions
79	Inventory System — Build Logic	SQL Interview Qs — Optimization
80	Interview Qs — Core Java	Interview Qs — Normalization/Design
81	Interview Qs — OOP	Mixed SQL Revision
82	Interview Qs — Collections	Resume & Portfolio Prep
83	Interview Qs — Exceptions/Threads	Resume & Portfolio Prep
84	Mock Interview 1 — Java Round	Mock Interview 1 — SQL Round

Week 13 (Days 85–90) — Final Week: Capstone & Graduation

Day	Java Session	SQL Session
85	Capstone — Architecture Planning	Capstone — Schema Finalize
86	Capstone — Core Entity Classes	Capstone — DAO-style CRUD Queries
87	Capstone — Service Layer + Exceptions	Capstone — Money/Order Transactions
88	Capstone — Polish & Finalize	Capstone — Reporting Queries + Indexes
89	Full Mock — Java + OOP/DSA	Full Mock — SQL + Design
90	Graduation — Resume/GitHub Checklist	Graduation — SQL Portfolio Checklist

4. Project Summary Cards

Console Calculator

Days 13–14

Skills: Methods, switch, SQL table design

Output: Working calculator + calc_history table

Student Management System

Days 25–27

Skills: Classes, CRUD, subqueries

Output: Repo with students/marks schema

Bank Management System

Days 41–44

Skills: Inheritance, interfaces, transactions/ACID

Output: Banking app + safe transfer queries

Hospital DB

Days 47–48

Skills: Collections, normalization

Output: Normalized hospital schema

Library Management System

Day 54

Skills: File handling, multithreading

Output: Borrow/return system

E-commerce DB

Days 53–61

Skills: Query optimization, ER diagrams

Output: Full e-commerce schema

Bank DB (SQL-focused)

Days 62–66

Skills: Execution plans, performance tuning

Output: Tuned reporting DB

Employee Management System

Day 69

Skills: Trees, hierarchical queries

Output: Org-chart style app

Inventory System

Days 78–79

Skills: Full OOP + business rules

Output: Stock/order management app

Capstone Backend

Days 85–90

Skills: Layered architecture, full SQL integration

Output: Portfolio-ready capstone + resume

5. Interview Preparation Kit

- ☐ Core Java, OOP, Collections, Exceptions/Threads — 15 Qs each (Days 80–83)
- ☐ SQL: Window Functions, Optimization, Normalization/Design (Days 78–81)
- ☐ Mock Interview 1 (Day 84) → Full Mock (Day 89)
- ☐ Resume & portfolio documentation (Days 82–83)

6. Progress Trackers

Metric	Target	Actual
Java problems solved	—	
SQL queries solved	—	
Projects completed	10	
Weekly checkpoints passed	13	
Monthly assessments passed	3	
Mock interview scores	—	

- ☐ Month 1 Complete
- ☐ Month 2 Complete
- ☐ Month 3 Complete
- ☐ Capstone Shipped
- ☐ Resume Finalized
- ☐ Applied to 5 Jobs